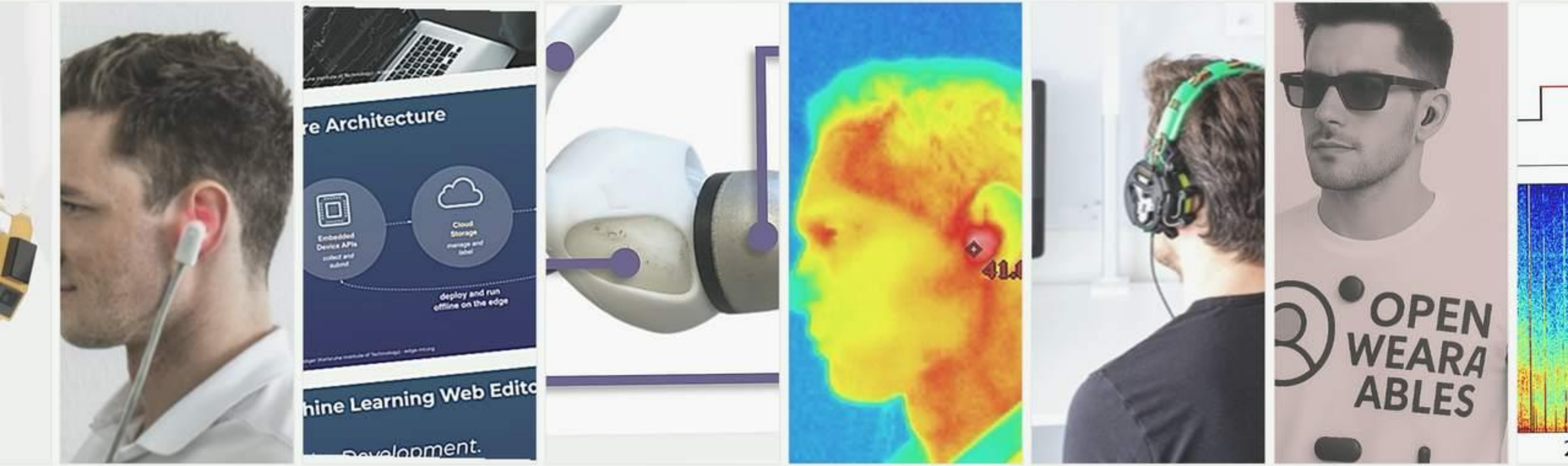


Human Computer Interaction SS26

Chapter 1

Intro

Michael Beigl, TECO

Definitions – Human-Computer Interaction

Wikipedia:

"Human-Computer interaction (HCI) is the study of interaction between people (users) and computers. It is an interdisciplinary subject, relating computer science with many other fields of study and research.
Interaction between users and computers occurs at the user interface (or simply interface), which includes both hardware (i.e. peripherals and other hardware) and software (for example determining which, and how information is presented to the user on a screen)."

SIG CHI:

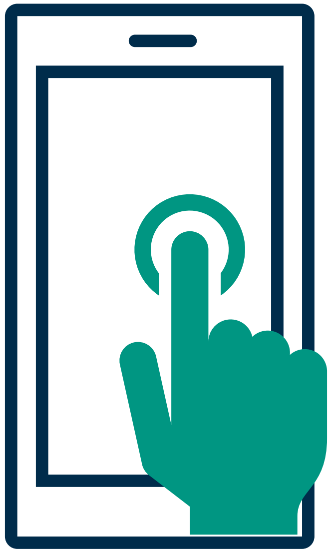
"Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them."

Lecture Overview (preliminiary)

Date / Day	Chapter	Topic
21 Apr 2026 (Tuesday)	Intro	
22 Apr 2026 (Wednesday)	HIP	Cognitive Foundations
28 Apr 2026 (Tuesday)	HIP	
29 Apr 2026 (Wednesday)	Exercise HIP / Lecture Integrated	
05 May 2026 (Tuesday)	Perception	
06 May 2026 (Wednesday)	Perception	
12 May 2026 (Tuesday)	Design Analysis	Design and Analysis
13 May 2026 (Wednesday)	Exercise Design Analysis and Perception	
19 May 2026 (Tuesday)	Design Analysis	
20 May 2026 (Wednesday)	Design	
26 May 2026 (Tuesday)	Pentecost	
27 May 2026 (Wednesday)	Pentecost	
02 Jun 2026 (Tuesday)	no lecture	
03 Jun 2026 (Wednesday)	Exercise Design Analysis	
09 Jun 2026 (Tuesday)	Design	
10 Jun 2026 (Wednesday)	Observation	Evaluation
16 Jun 2026 (Tuesday)	Observation	
17 Jun 2026 (Wednesday)	Exercise Observation	
23 Jun 2026 (Tuesday)	Interface and Interaction	Technology and Interaction
24 Jun 2026 (Wednesday)	Interfact and Interaction	
30 Jun 2026 (Tuesday)	Exercise Interface	

Human-Computer Interaction

What do **we design:** **systems**
- Devices
- Components
- Products
- Software that process information in the context of



interactive systems
- respond to people’s actions
- mediate information in a way people can perceive

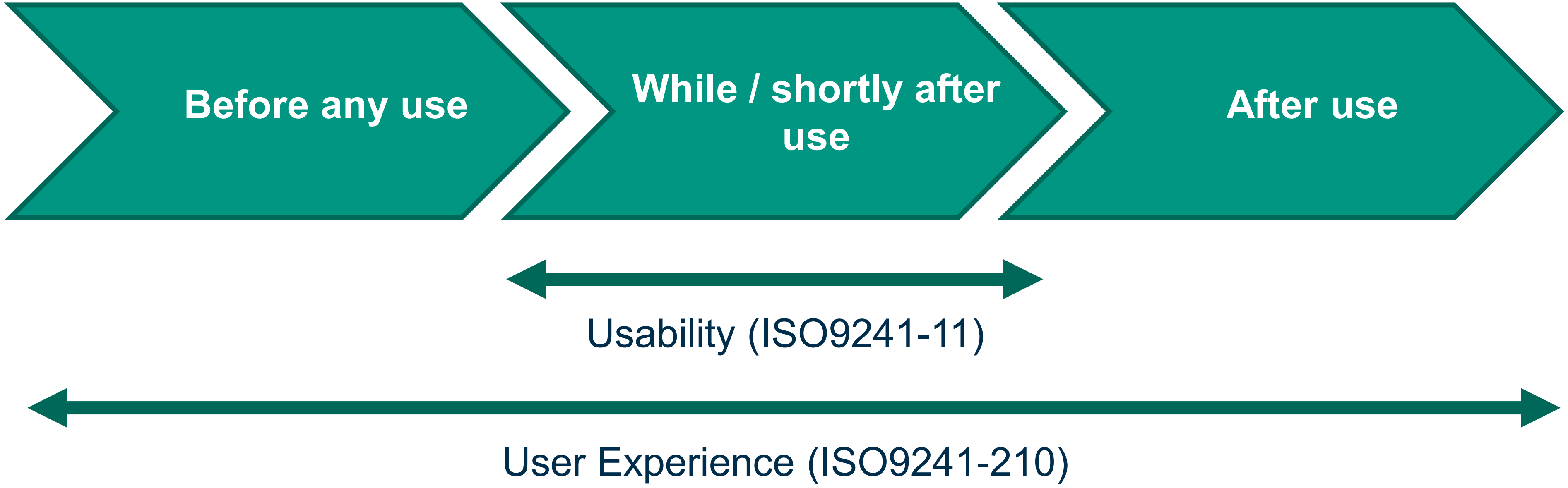
→ everything you recognize (see, touch, feel,..) and act to exchange information with a technical system

Why YOU should care about HCI

The user experience **is important** …

https://www.youtube.com/watch?v=HtTUsOKjWyQ

Usability vs. User Experience



Conversational AI as a new interface paradigm

- Natural language becomes a general input channel for many tasks.
- Chat interfaces lower the entry barrier for search, brainstorming, explanation, coding and drafting.
- The interface is increasingly multimodal: text, voice, image and context are combined.
- For HCI, this means interaction design moves from fixed commands toward dialogue, repair and negotiation.

What becomes an HCI question?

- How much state is visible?
- How are uncertainty and errors communicated?
- How much memory, initiative and control should the system have?

Lecture takeaway:

Chat is not only a new tool; it is a new interface class.

Sources: Namvarpour & Razi, The Art of Talking Machines (CUI 2025); Luera et al., Survey of UI Design and Interaction Techniques in GenAI Applications (2024).

Why chat matters - and where it breaks down

Strengths of chat-based interfaces

- Low threshold and flexible input
- Good for exploration and underspecified problems
- Useful when users do not know the exact command or menu path
- Can explain, summarize, reformulate and personalize

Typical weaknesses

- High formulation effort and hidden system state
- Weak support for precise comparison and direct manipulation
- Risk of overtrust because language sounds confident
- Long workflows become difficult without visible structure

Design implication: combine chat with visible controls, history and direct manipulation.

Sources: DirectGPT (CHI 2024); Navigating the Unknown / CARE (IUI 2025); GenerativeGUI (CHI EA 2025); Namvarpour & Razi (CUI 2025).

Activities and Technologies

Figure 2.1 Activities and technologies.

Source: after Carroll (2002), Figure 3.1, p. 68., graphically modified

History of Human-Computer Interaction

Interactive Systems Timeline - additions since 2022

2022
ChatGPT

Natural-language entry point for many everyday tasks

2024
multimodal copilots

Text + image + voice + context

2024/25
GUI agents

Systems begin to act inside graphical user interfaces

2025
dynamic GUI

Chat interfaces can generate task-specific controls

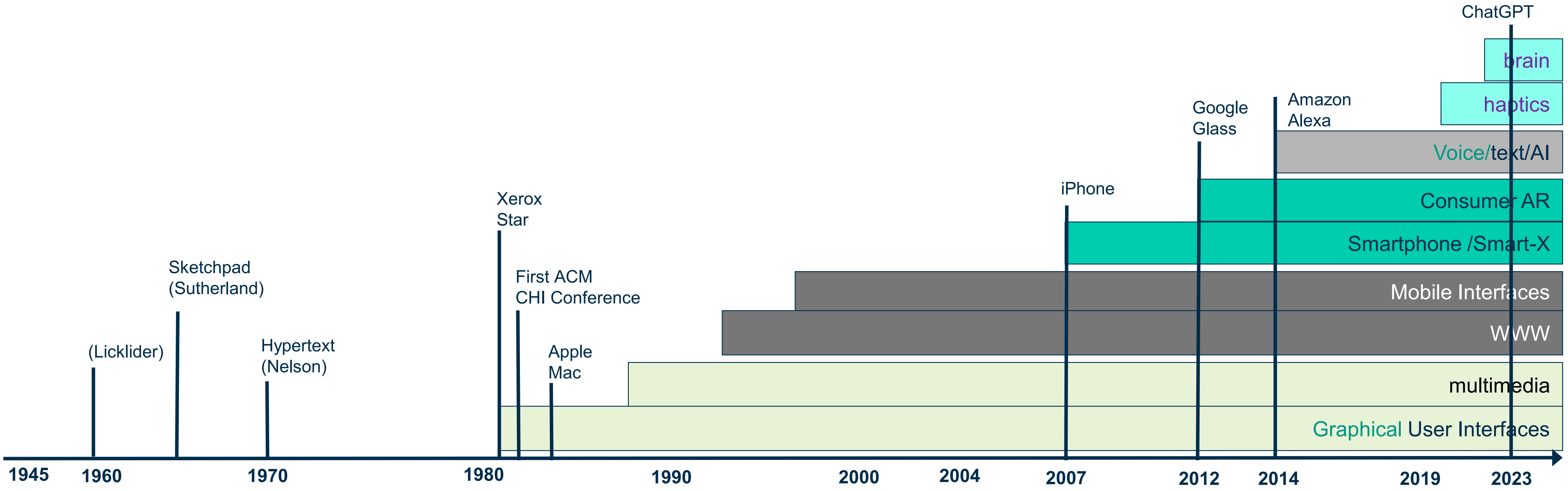
2026
collaborative agent presence

AI participation becomes a group interaction question

Post-GUI era now includes conversational, multimodal and agentic interface layers.

Sources (adopted): Luera et al. (2024); GUI Agents: A Survey (2024/25); Johnson et al., Interface-Driven Social Prominence in Collaborative GenAI Systems (CHI 2026).

Interactive Systems Timeline



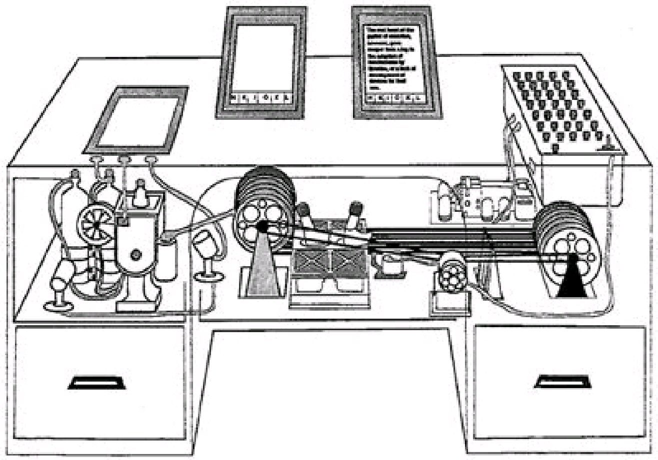
Designing Interactive Systems; D. Benyon, P. Turner and S. Turner; 2005, Pearson, p. 17 ; Extended and design changed by me

Post-GUI era:

- Touch, Virtual Reality (VR), Mixed Reality (MR), Smart-X, Voice
- Haptic User Interfaces

Vannevar Bush

- „**As we may think**“ 1945, based an article written in 1939
- V. Bush was since advisor of the US gov. in WWII and after
- Among others, he created the idea of the **Memex**
- Professor of Claude Shannon
- This is perceived as a first expression of the idea of
  - **Desktop Computing**
  - **Internet and Hypertext/Web Technology**



Main claim: As machines have taken away the burden for performing mechanical, stupid activity, novel technology will taken away the burden for information storing/retrieval
Memex stores/linkes books, etc.

X-Y position indicator for a display system, 1967

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First Mouse (Douglas Engelbart)

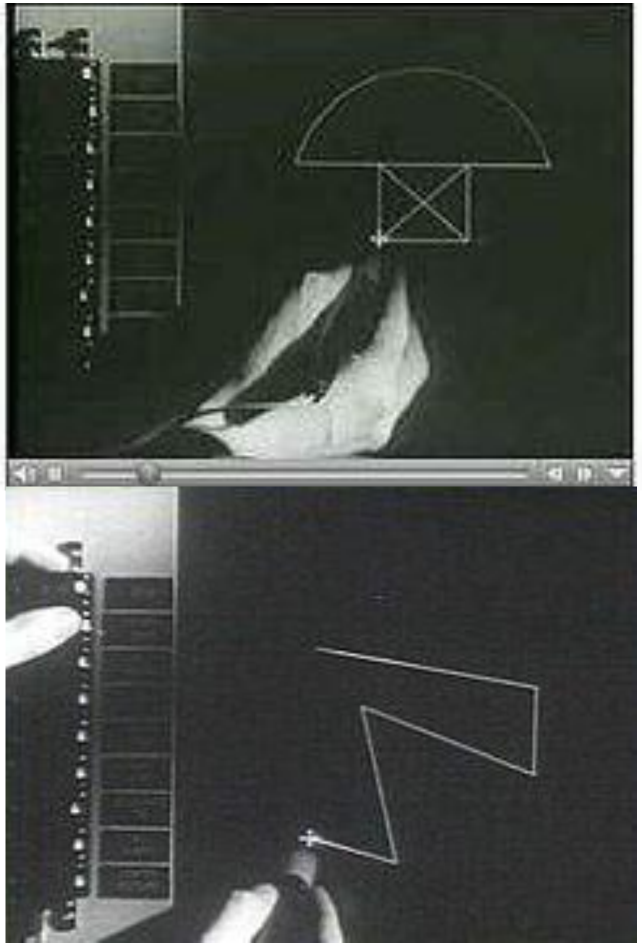
Douglas Engelbart

https://www.youtube.com/watch?v=B6rKUf9DWRI

SketchPad: First CAD Program, 1963

Breakthrough for computer graphics

First complete GUI-based program



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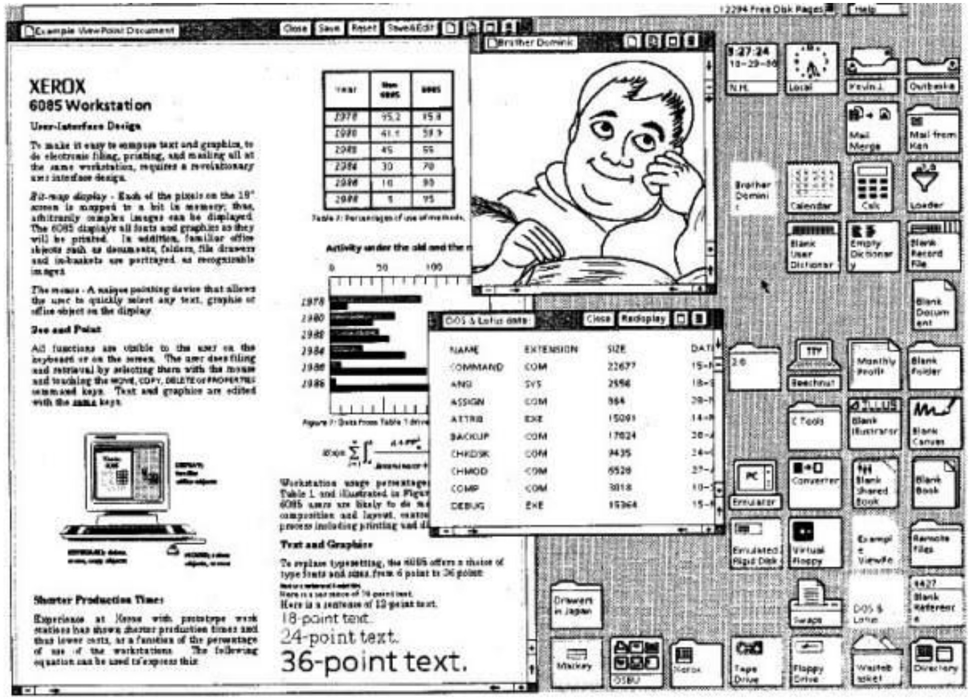
Idea of objects and instances (e.g. shapes)

Xerox Star, 1977



bitmapped display, graphical user interface, icons, folders, mouse!

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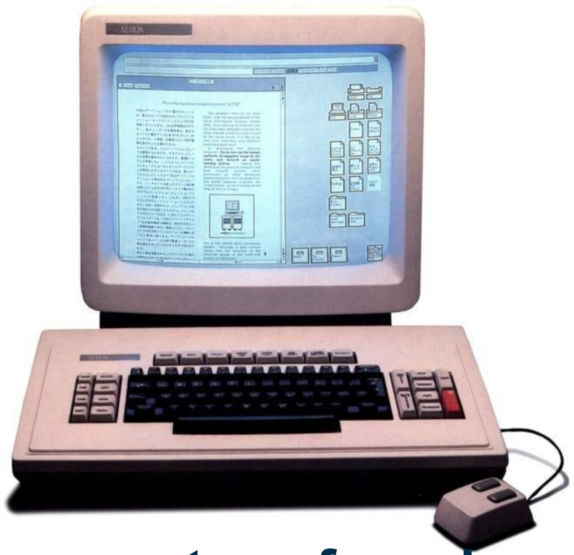

ancestor of modern personal computers!

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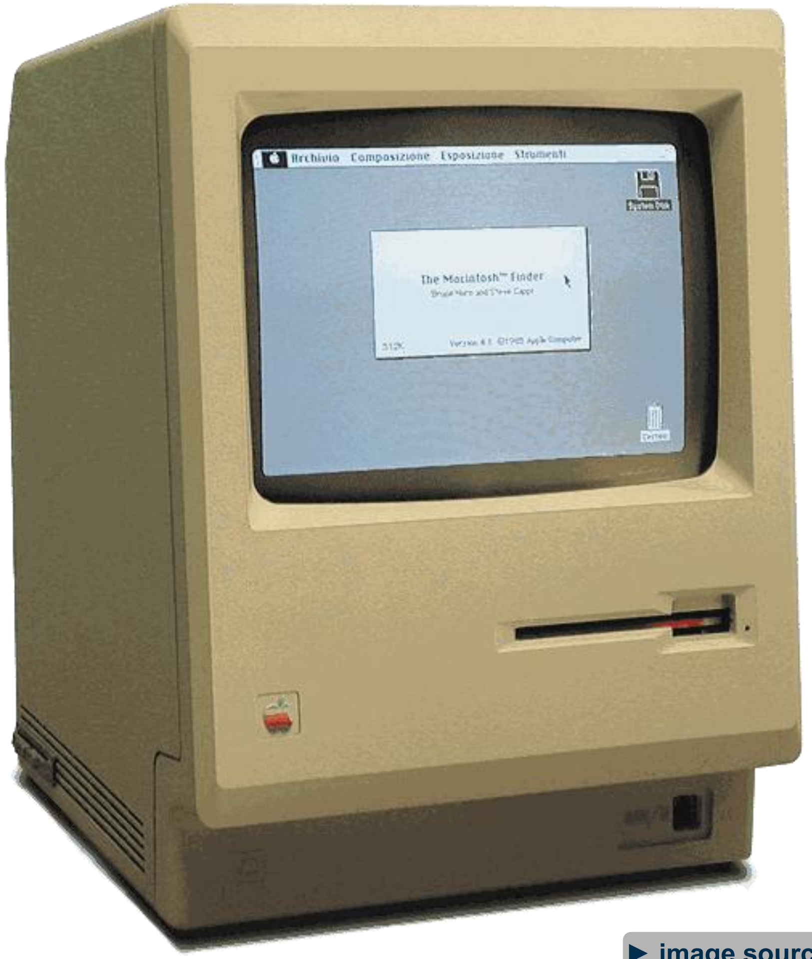
Visible Content
WYSIWYG: What you see is what you get

Apple Mac / Macintosh, 1984

color display, graphical user interface (GUI) and mouse, affordable



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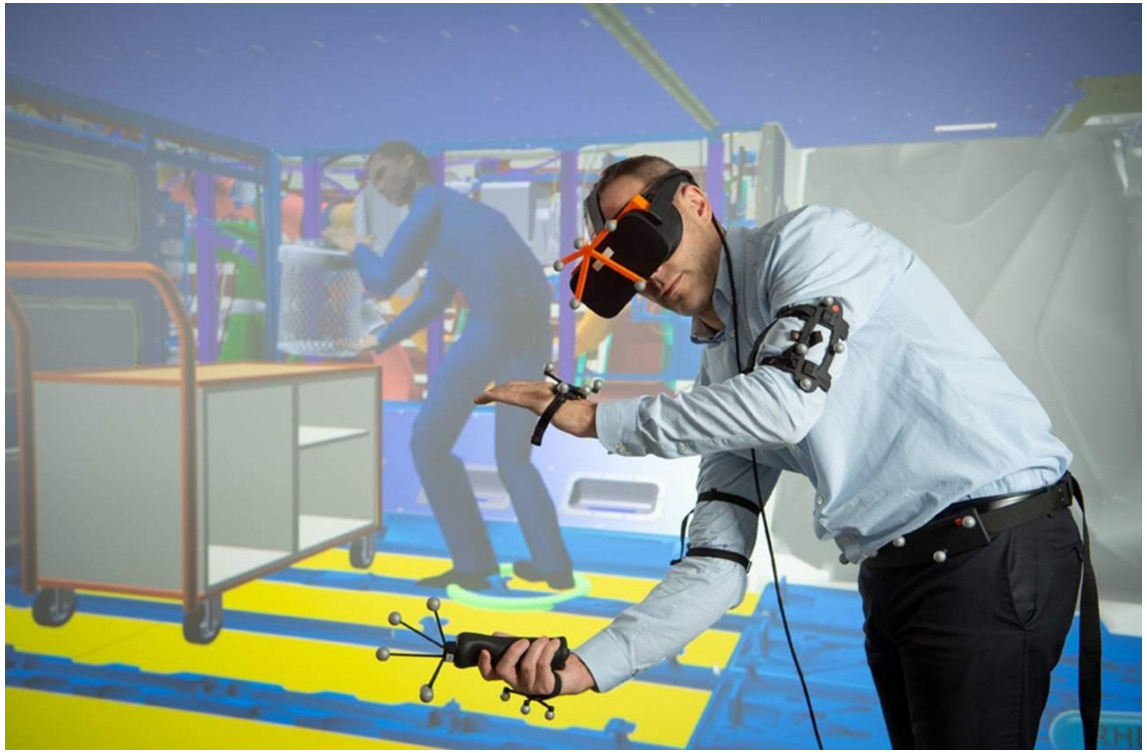
Virtual and Augmented / Mixed Reality



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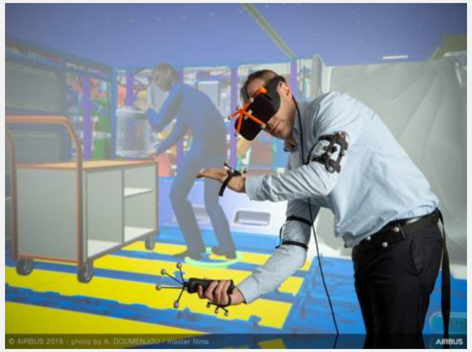
„Overlay" on real world

The world is the interface!

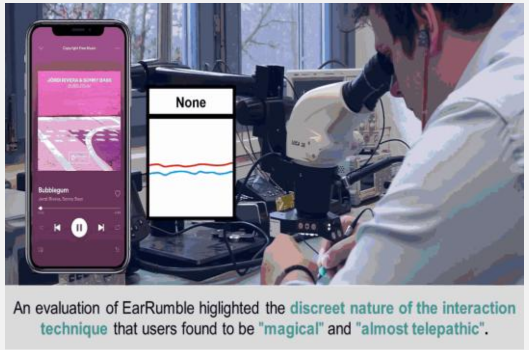
Proliferation now includes AI interaction layers

Smartphone



AR / XR

Brain / biosignals

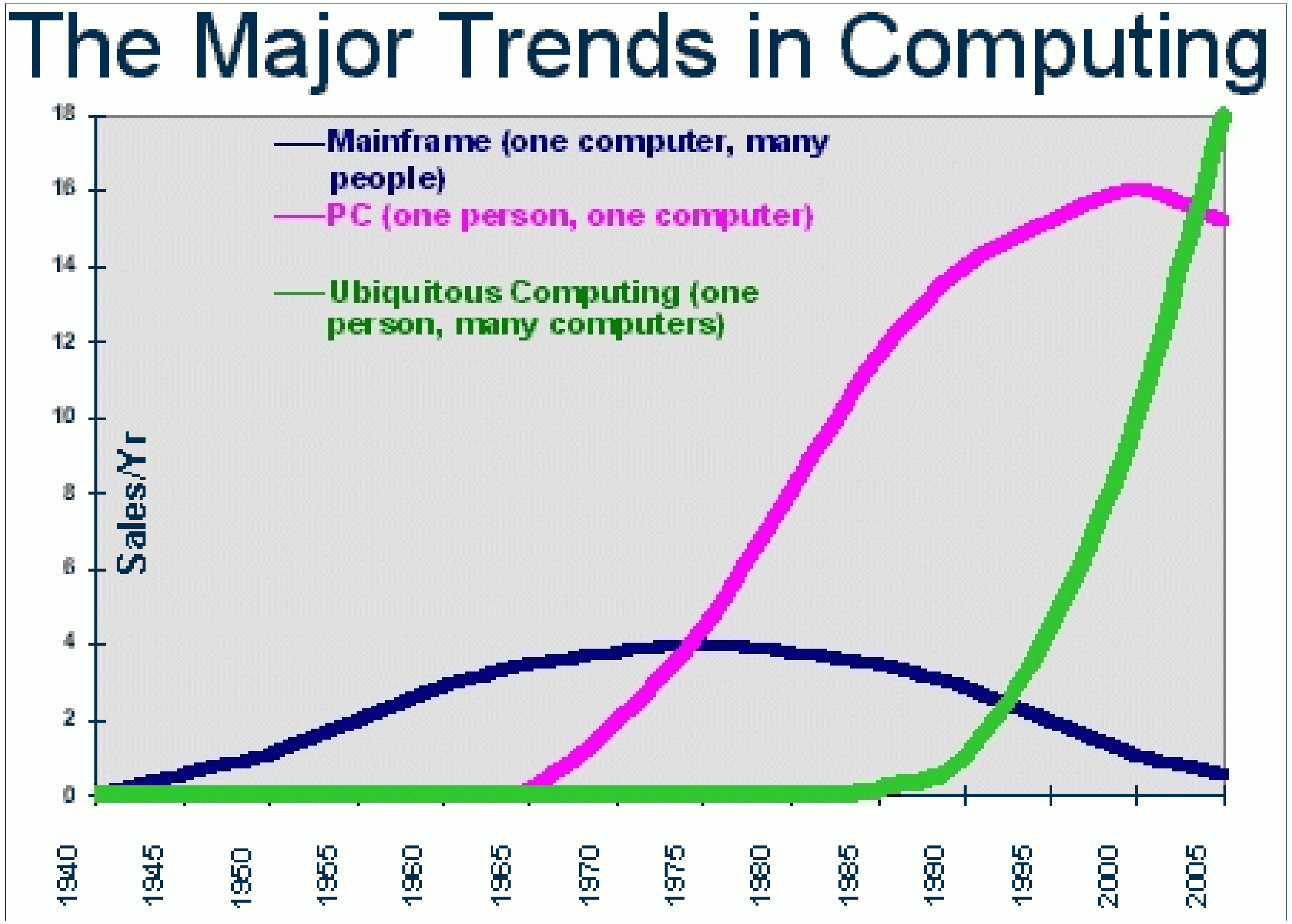


Earables / subtle input

The important change is not only more devices; it is that many devices now share conversational and AI-mediated interaction patterns.

Sources: Original lecture examples; Luera et al. (2024); Namvarpour & Razi (2025).

Proliferation of interactive systems



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Touch user interaction

Direct interaction with the screen and it‘s elements

„the content is the interface“ (Edward Tufte)

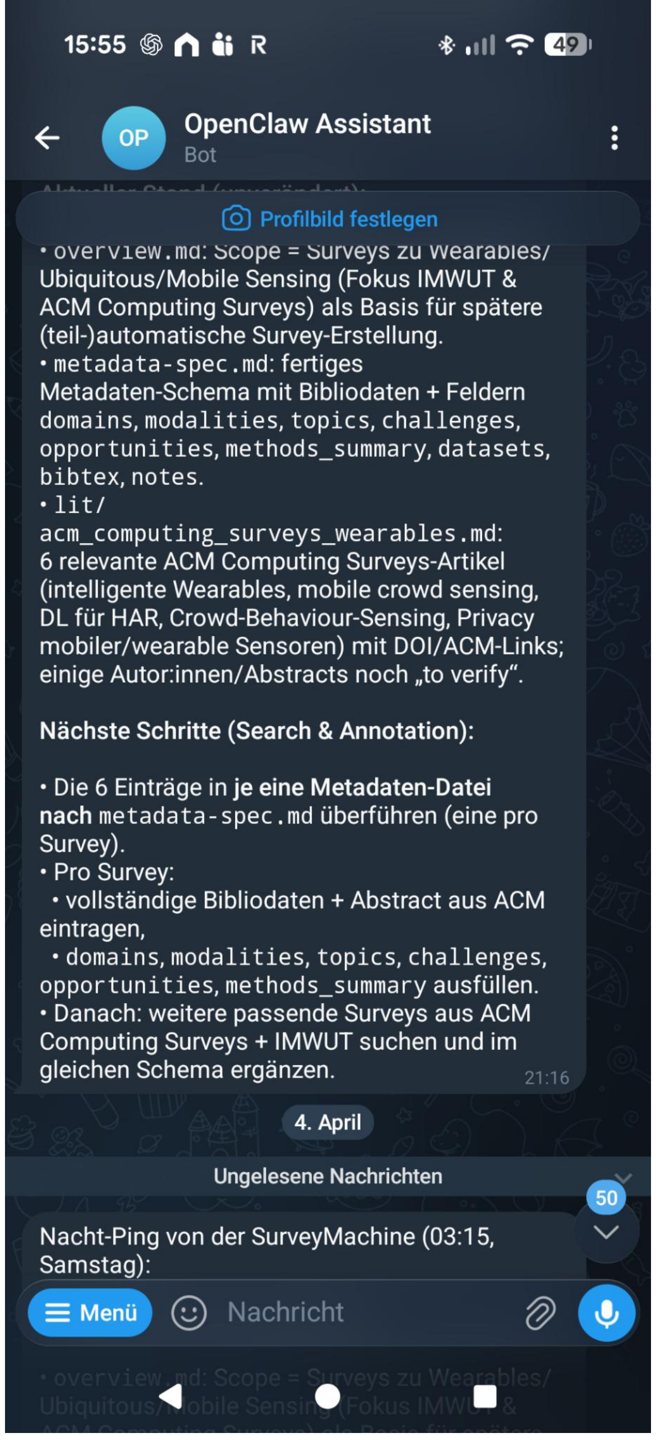
▶ image source

Conversational User Interfaces in HCI

- Conversational user interfaces (CUIs) are an umbrella term for interfaces that enable dialogue through natural language.
- CUIs include three common forms:
  - text-based interfaces, e.g. chatbots
  - voice-based interfaces, e.g. voice user interfaces / voice assistants
  - multimodal or embodied conversational interfaces

UI agents and mixed initiative

- UI agents use AI (often) language models to operate existing interfaces.
- They allow automation for multi-step tasks in web, desktop and mobile systems.
- In HCI terms, this shifts interaction from "doing" to "delegating".
- The central design question becomes: when should the system act, ask, stop or explain?



Brain and Haptic Interfaces

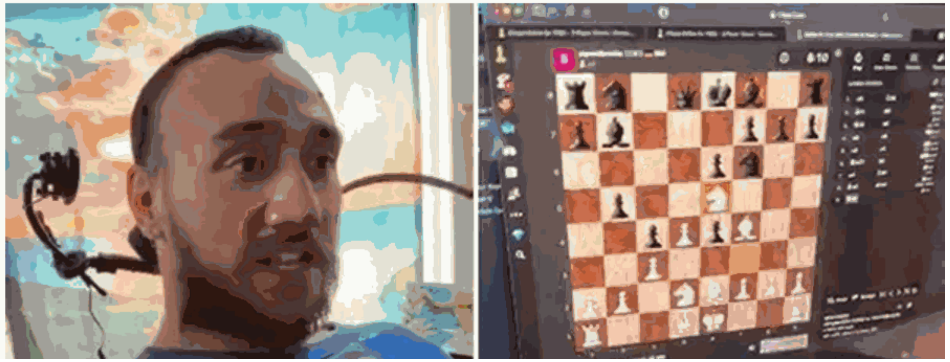
Invasive and non-invasive methods to directly access
- Input: brain waves
- Output: haptic sensations

Both rely on neurological adaption

Currently for specific applications
- Medical, sports, industry

Center for Sensorimotor Neural Engineering (CSNE), Photo by Mark Stone,

teco.kit.edu

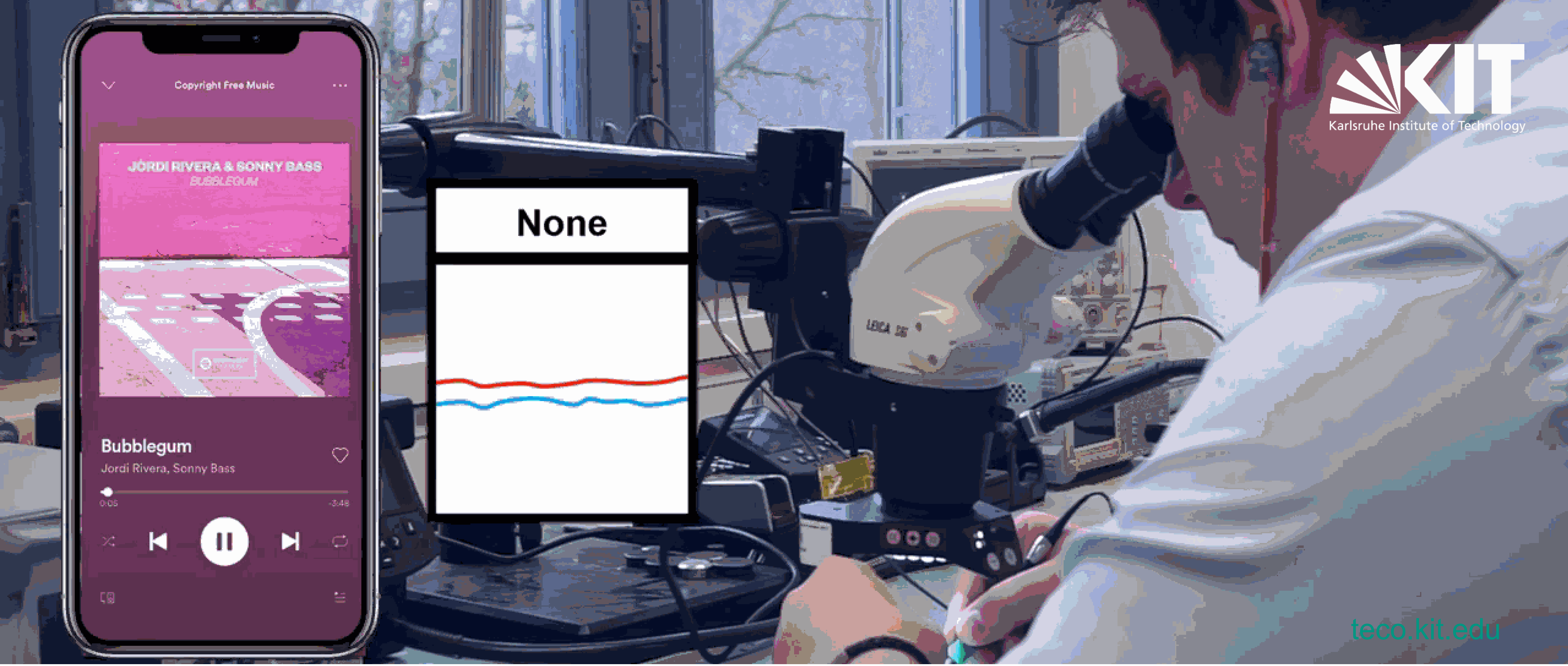


Neuralink.com

Some recent references on AI interfaces in HCI

- Pang et al. (2025). Understanding the LLM-ification of CHI.
- Namvarpour & Razi (2025). The Art of Talking Machines.
- Masson et al. (2024). DirectGPT.
- Peng et al. (2025). Navigating the Unknown / CARE.
- Hojo et al. (2025). GenerativeGUI.
- Nguyen et al. (2024/25). GUI Agents: A Survey.
- Johnson et al. (2026). Interface-Driven Social Prominence in Collaborative GenAI Systems.

(all ACM CHI or IUI)

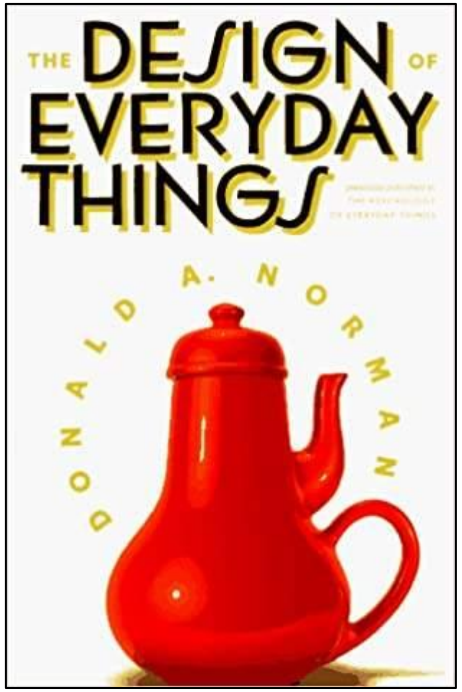


In an application study with 8 participants, the new EarRumble technique was described as “magical” and “almost telepathic”.

EarRumble: Discreet Hands- and Eyes-Free Input by Voluntary Tensor Tympani Muscle Contraction. T. Röddiger, C. Clarke, D. Wolffram, M. Budde, M. Beigl. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems

Some books

The Design of Everyday Things (Don Norman), The Invisible Computer, The Visual Display of Quantitative Information (Edward Tufte)



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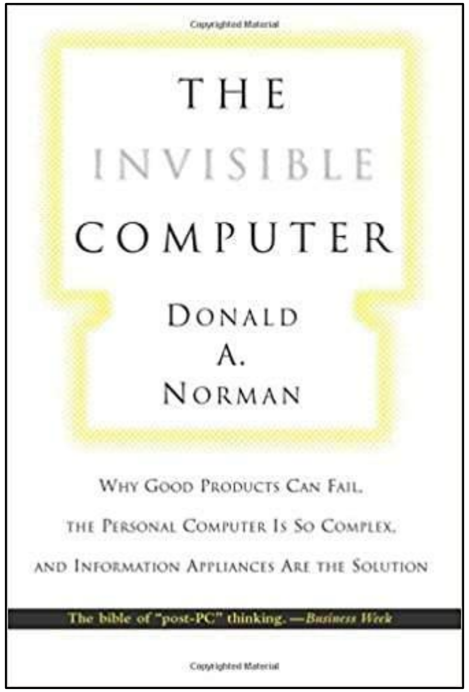

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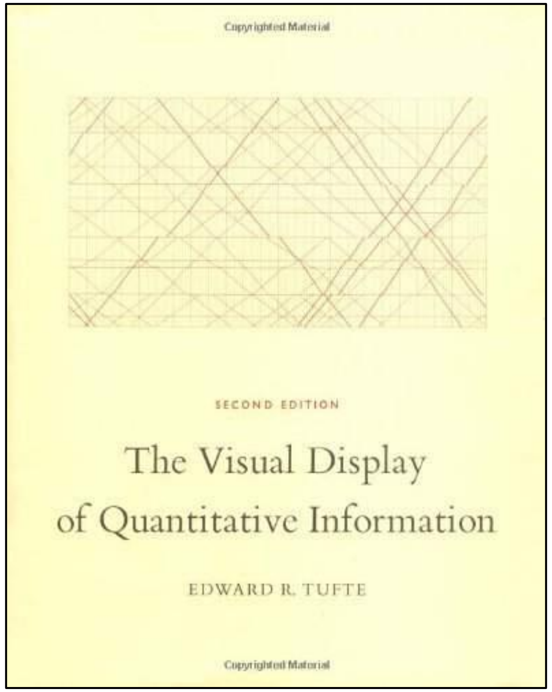

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